TITLE:

BUILD PLANET ZERO HOUSING REGENERATION #2

Prepared for:

BUILD PLANET ZERO LIMITED

Prepared by:

Muktha Nanaiah Achakalera

Abin S J Manju Raj

Johnson Pius

Sunday Ononfa

Okey Ilobi

Abstract

This technical report explores sustainable construction materials and methods for addressing modern housing and infrastructure challenges, focusing on the Wallsend region. It evaluates vendors and materials, including Structural Insulated Panels (SIPs), self-healing concrete, Cross-Laminated Timber (CLT), and hempcrete, emphasizing their environmental benefits, durability, and suitability for various applications.

The report highlights the integration of advanced building solutions, such as renewable energy systems, rainwater harvesting, and eco-friendly materials, to achieve net-zero targets. Vendors like Maitland Cool Rooms and Hemsec Manufacturing provide SIPs tailored for residential and commercial use, while UK Hempcrete specializes in bio-based materials for stable, energy-efficient structures. Concrete solutions from Tarmac and Breedon ensure sustainable and long-lasting construction, complemented by timber offerings from BSW Timber and Quay Timber.

By leveraging innovative construction practices, compliance with regulations like BREEAM and SuDS, and government incentives for renewable energy and water conservation, this report outlines a pathway to creating affordable, sustainable, and high-quality living spaces. The methodology combines historical and site-specific considerations, stakeholder collaboration, and best practices from case studies to guide the transition to eco-friendly urban development. This work serves as a blueprint for integrating sustainability into construction while addressing social, economic, and environmental needs.

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1. Introduction

1.1 Background: Overview of the Construction Project and Its Purpose

The project addresses the urgent UK housing crisis, characterized by soaring rental and energy costs, which have left many struggling to access affordable, high-quality housing. In response, the private landlord has committed to transforming the Wallsend site into eco-friendly homes. The previous plan, focused on retail units and apartments, did not meet the growing need for affordable and sustainable living. Collaborating with Build Planet Zero, a net-zero consultancy, the project seeks to re-submit a planning application to develop a more sustainable and community-oriented housing scheme.

1.2 Objectives: Specific Goals of the Project

- Design a **Net-Zero-Ready housing scheme**, powered entirely by renewable energy.
- Provide **eco-friendly rental homes** and communal spaces at affordable costs, targeting low-income residents.
- Ensure all homes are **highly energy-efficient**, incorporating renewable energy and low-carbon building technologies.
- Adopt a **life-cycle and user-oriented approach** to sustainable design.
- Promote **happy, healthy living environments** while reducing environmental impact.
- Set a benchmark for **private-led low-carbon housing** in the North-East of England.

1.3 Scope:

- **Site Area**: 655m^2, with one remaining building from a previous demolition.
- **Redevelopment Plan**: Re-submit a planning application focused on sustainability, replacing the prior plan.
- **Project Deliverables**: Development of Net-Zero-Ready homes, communal spaces, and relevant facilities.
- **Stakeholder Roles**: Build Planet Zero acts as the Sustainability and Net Zero integrator, working with various experts, partners, and stakeholders, including the private landlord and architecture and built environment (ABE) students from Northumbria University.

2. Project details

2.1 Project Description

The project is located on High Street, Wallsend, in North Tyneside, a short commuting distance from Newcastle city center. The site spans 655m² and was previously occupied by four two-storey, mid-terrace mixed-use properties with retail units on the ground floor and residential apartments above. Following the demolition of three units in 2015, one unit remains on-site. The redevelopment aims to transform the site into a **Net-Zero-Ready housing scheme**, incorporating eco-friendly rental homes and communal spaces powered by renewable energy. The project will prioritize sustainable design, affordability, and quality living for low-income individuals.

2.2 Site description

- **Location**: High Street, Wallsend, a well-connected area near Newcastle city center.
- **Topography**: The site is relatively flat, suitable for low-rise construction.
- **Climate**: The region experiences a temperate maritime climate, influencing the design and energy efficiency measures for the development.
- **Existing Infrastructure**:
  - One remaining mid-terrace unit from the original four mixed-use properties.
  - Space allocated for parking (8 spaces), refuse stores, and other facilities.
  - The area is accessible and well-suited for residential development with adequate connectivity.

2.3 Project Team

- **Private Landlord**: Owner of the site and initiator of the redevelopment plan.
- **Build Planet Zero**: Sustainability and Net Zero integrator, responsible for:
  - Developing sustainability strategies and eco-designs.
  - Incorporating low-carbon energy systems into the project.
  - Coordinating stakeholders, including researchers, architects, engineers, and operators, to ensure a holistic approach.
- **Engineering and Built Environment Students (Northumbria University)**:
  - Assisting with design and implementation tasks under supervision.
  - Supporting collaboration between teams and contributing to project deliverables.

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3. Methodology

3.1 Site and Historical Context Analysis

The project site is situated in Wallsend, a town with significant historical and industrial importance:

- Roman Heritage: The town served as the eastern endpoint of Hadrian's Wall, contributing to its international recognition as part of the UNESCO World Heritage Site.
- Industrial Growth: Wallsend was historically renowned for its shipbuilding and pioneering role in electrical power generation along the River Tyne during the 19th and early 20th centuries.
- Modern Developments: While the decline of shipbuilding in the late 20th century led to economic challenges, recent opportunities in advanced manufacturing, particularly offshore energy, have revitalized the region.

This historical and industrial analysis helps contextualize the need for regeneration and sustainable housing development that respects Wallsend's heritage while addressing contemporary challenges.

3.2 Strategic Context for Regeneration

The project aligns with the strategic goals outlined in the Our North Tyneside Plan and the Council's regeneration strategy:

- Improving Housing Quality: Tackling poor-quality private housing and providing affordable, sustainable homes.
- Revitalizing the Town Centre: Enhancing public spaces, improving safety, and supporting local businesses.
- Connecting Residents to Opportunities: Providing access to good jobs through improved transport infrastructure and adult education.

These policy priorities frame the methodology for housing development and refurbishment in Wallsend.

Table 4: Phased Housing Requirement

	Phase 1 – 2011/12 to 2015/16	Phase 2 – 2016/17 to 2020/21	Phase 3 – 2021/22 to 2025/26	Phase 4 – 2026/27 to 2030/31	Phase 5 – 2032	Total Requirement
Total	2,755	3,700	4,690	4,540	908	16,593
Per Annum	551	740	938	908	908	790

Fig 1. Housing requirement from north Tyneside local plan

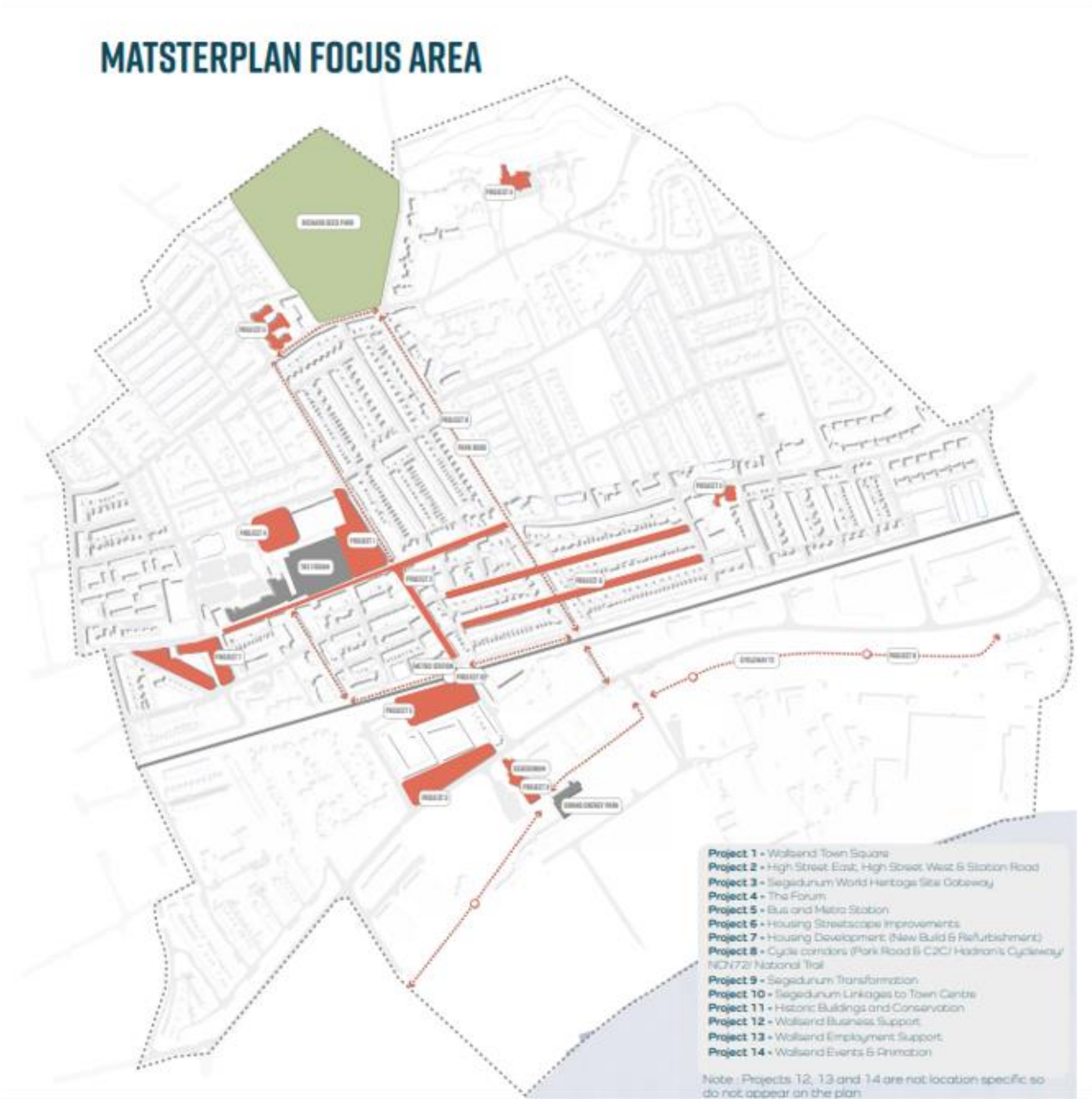

Fig 2. Wallsend Master plan

3.3 Planning and Regulatory Framework

1. National Planning Policy Framework (NPPF):
    - Ensures developments align with sustainable and community-centered goals.
    - Key considerations include environmental impact, heritage conservation, and community engagement.
2. North Tyneside Local Plan:
    - Defines land use, housing allocations, and conservation guidelines.
    - Development near heritage sites (e.g., Hadrian's Wall) requires adherence to stricter regulations.

3. Building Regulations:

    - Structural and energy efficiency standards (e.g., Part L for energy conservation) govern construction to ensure safety and sustainability.

4. Permitted Development Rights:

    - Small extensions and modifications may not require planning permissions unless located in designated conservation areas.

Table 1 - Minimum gross internal floor areas and storage (m^2)

Number of bedrooms(b)	Number of bed spaces (persons)	1 storey dwellings	2 storey dwellings	3 storey dwellings	Built-in storage
	1p	39 (37) *			1.0
1b	2p	50	58		1.5
	3p	61	70		
2b	4p	70	79		2.0
	4p	74	84	90	
3b	5p	86	93	99	2.5
	6p	95	102	108	
	5p	90	97	103	
	6p	99	106	112	
4b	7p	108	115	121	3.0
	8p	117	124	130	
	6p	103	110	116	
5b	7p	112	119	125	3.5
	8p	121	128	134	
	7p	116	123	129	
6b	8p	125	132	138	4.0

Table 1. required floor area by the Design Quality Supplementary Planning Document

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3.4 Demographic Analysis

1. Data Sources

    - Strategic Housing Market Assessment (SHMA)
    - Office for National Statistics (ONS) Census Population Data
    - Local household survey (2021)

2. Demographic Evaluation Parameters

Key demographic factors examined:

- Population composition
- Housing stock characteristics
- Residential tenure types
- Economic indicators
- Community needs assessment

Demographics

Current population: The latest population estimate for North Tyneside is **210,487 (2022)** of which **108,184** - female **102,303** - male

Median age of North Tyneside residents (2022)
43.2

Broad Age Group	Female	Male	Population
0 - 15	17,967	19,379	37,346
16 to 64	66,158	63,101	129,259
over 65 (Elderly)	24,059	19,823	43,882
Total	**108,184**	**102,303**	**210,487**

Estimated North Tyneside population

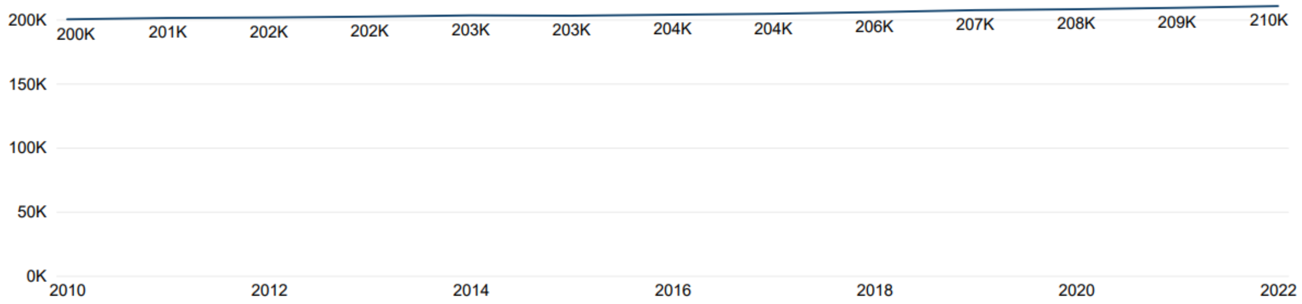
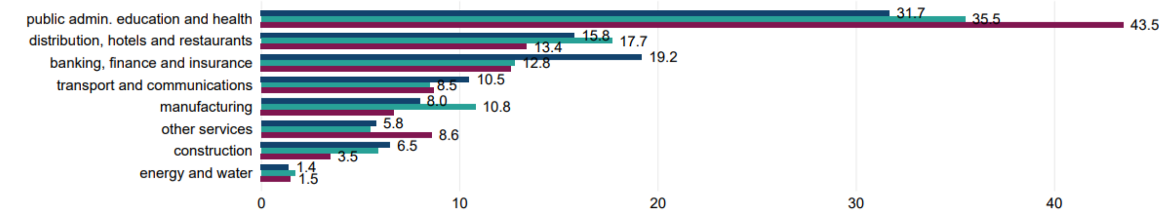


Fig 3

Employment

Employment of residents by industry (%)
October 2022-September 2023
● England ● North East ● North Tyneside



Employment of residents by occupation (%)
October 2022-September 2023
● England ● North East ● North Tyneside

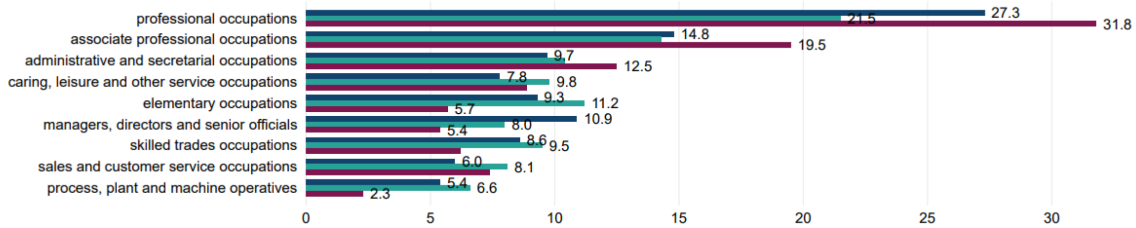
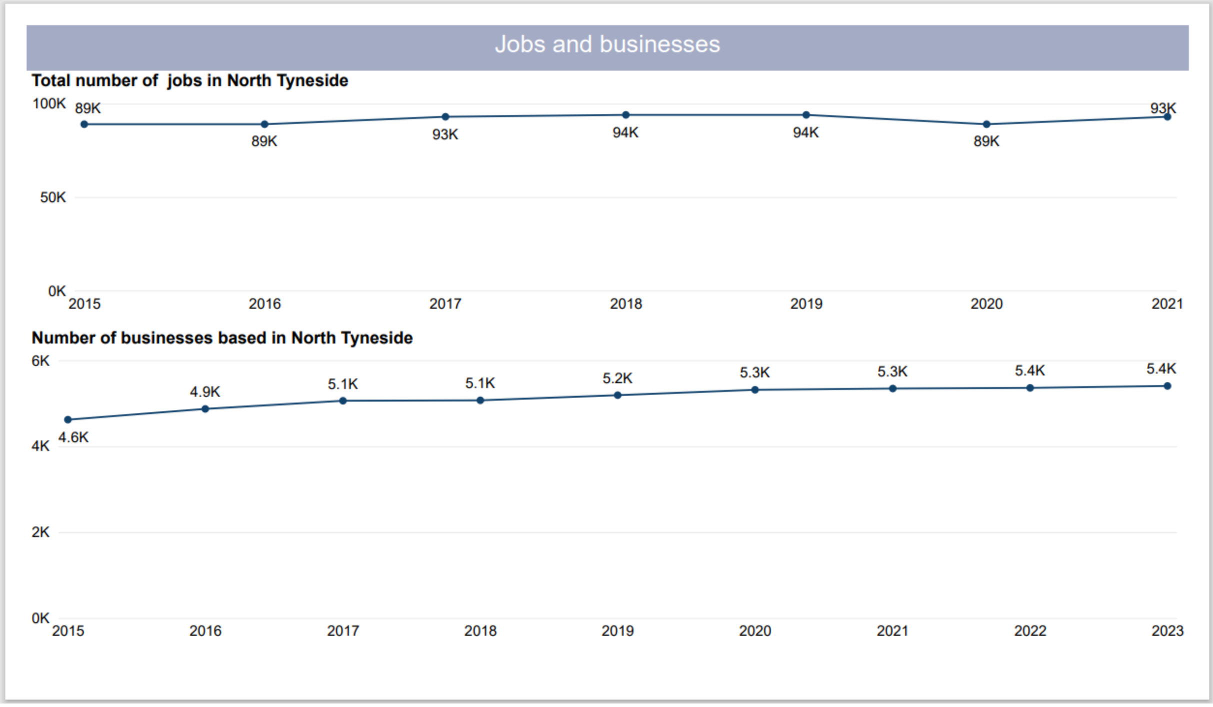
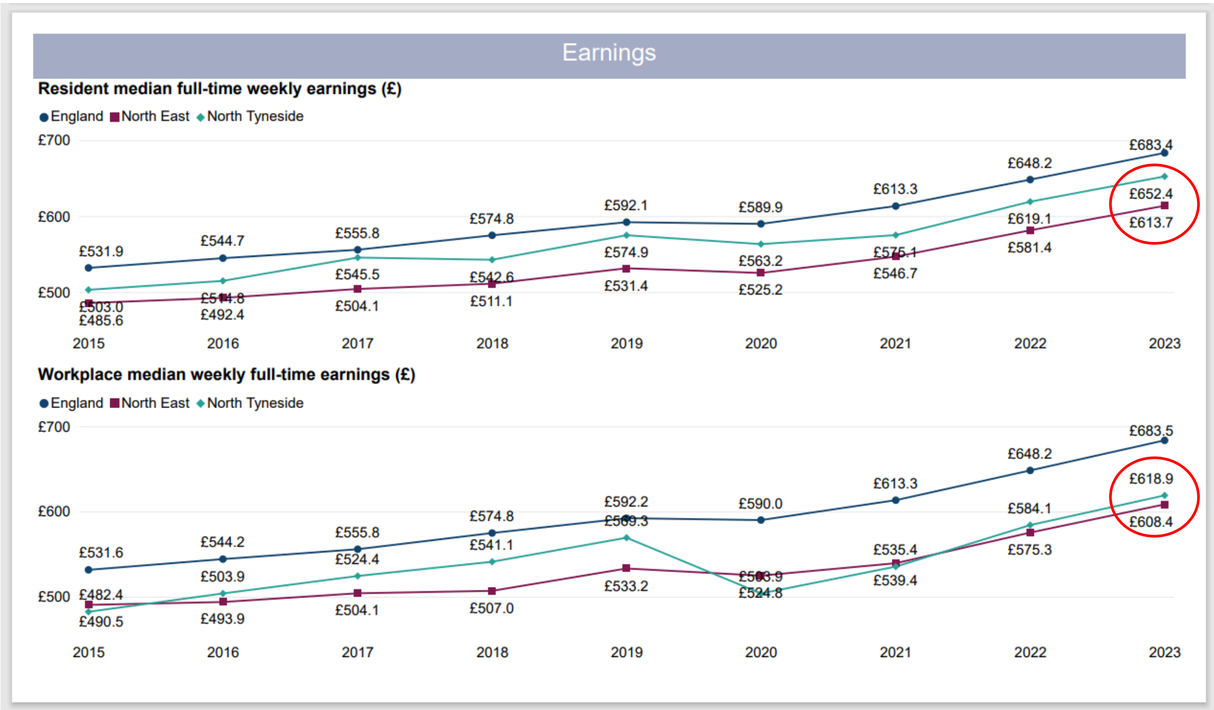


Fig 4



Jobs and businesses
Total number of jobs in North Tyneside
100K
50K
0K
89K
89K
93K
94K
94K
89K
93K
2015
2016
2017
2018
2019
2020
2021
Number of businesses based in North Tyneside
6K
4K
2K
0K
4.6K
4.9K
5.1K
5.1K
5.2K
5.3K
5.3K
5.4K
5.4K
2015
2016
2017
2018
2019
2020
2021
2022
2023

Fig 5



Earnings
Resident median full-time weekly earnings (£)
England North East North Tyneside
£700
£600
£500
£531.9
£544.7
£555.8
£574.8
£592.1
£589.9
£613.3
£648.2
£683.4
£652.4
£613.7
£619.1
£581.4
£574.9
£563.2
£545.5
£542.6
£531.4
£525.2
£546.7
£504.1
£511.1
£492.4
£485.6
2015
2016
2017
2018
2019
2020
2021
2022
2023
Workplace median weekly full-time earnings (£)
England North East North Tyneside
£700
£600
£500
£531.6
£544.2
£555.8
£574.8
£592.2
£590.0
£613.3
£648.2
£683.5
£618.9
£608.4
£584.1
£575.3
£541.1
£524.4
£503.9
£535.4
£539.4
£533.2
£524.8
£482.4
£490.5
£493.9
£504.1
£507.0
2015
2016
2017
2018
2019
2020
2021
2022
2023

Fig 6

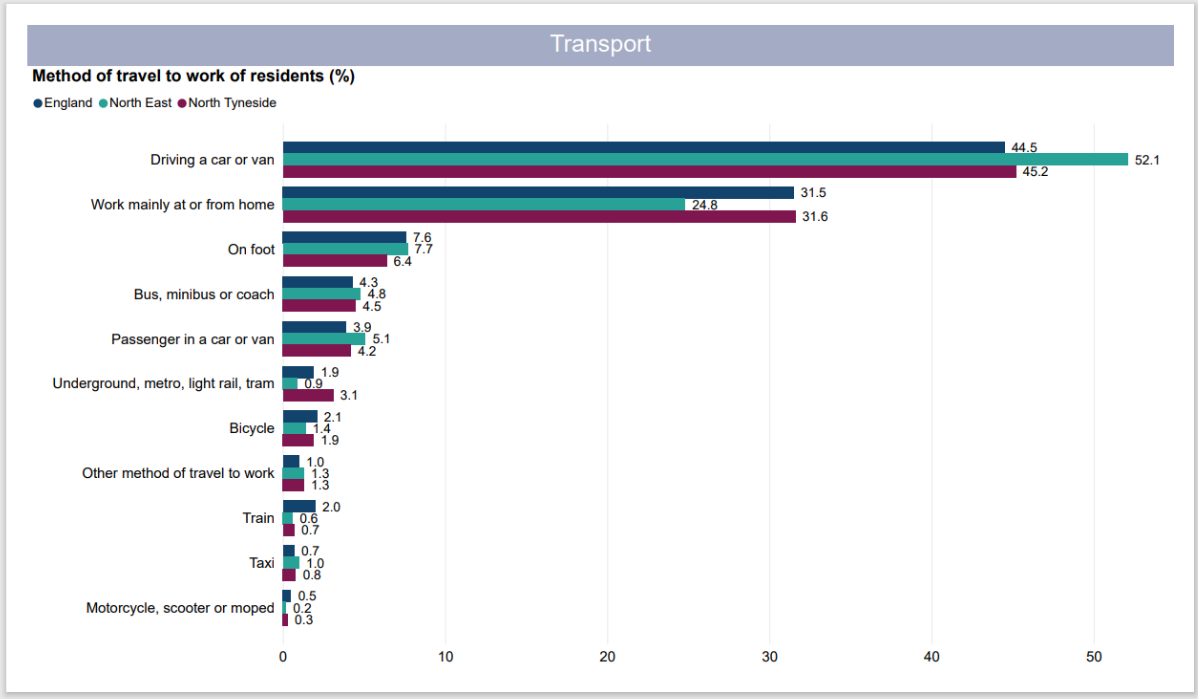


Fig 7

Property Rents in Wallsend by Number of Bedrooms

	No. of properties	Average rent	Median rent
One bedroom	1	£495 pcm	£495 pcm
Two bedrooms	13	£672 pcm	£650 pcm
Three bedrooms	9	£1,008 pcm	£850 pcm
Four bedrooms	0	-	-
Five bedrooms	0	-	-

Fig 8

3.5 Design and Sustainability Integration

The project incorporates sustainable design principles based on the Design Quality Supplementary Planning Document (SPD) and best practices from exemplary UK housing projects:

1. Architectural Design
   - Develop designs that harmonize with the local historical and architectural context.
   - Methodology: Use designs that blend into the existing urban fabric while modernizing layouts for functionality and energy efficiency.
2. Sustainability Measures
   - Incorporate low-carbon technologies, energy-efficient building materials, and renewable energy systems (e.g., solar panels).
   - Methodology: Apply life-cycle assessment techniques to optimize building performance and minimize environmental impact.
3. Community Spaces
   - Integrate accessible, safe, and attractive communal spaces to encourage community interaction and improve residents' quality of life.
   - Methodology: Include shared green spaces, safe pathways, and energy-efficient communal facilities in the design.

3.6 Comparative Analysis of Best Practices

To ensure a robust housing strategy, the project methodology includes studying and incorporating lessons from successful UK housing projects:

Project Name	Key Features	Methodological Insights
Goldsmith Street	Passivhaus-certified, energy-efficient, social housing	Design for sustainability reduces long-term costs while maintaining affordability.
Marmalade Lane	Cohousing model with shared spaces	Community-driven design reduces social isolation and improves affordability.
New Ground Cohousing	Housing tailored for older adults	Age-appropriate housing ensures adaptability and communal support.
Thamesmead Regeneration	Modular building with green infrastructure	Scalability and advanced methods optimize large-scale housing delivery.
Home for Heroes	Prefabricated, affordable homes for ex-service personnel	Purpose-built solutions meet specific community needs affordably.

Table 2

3.7 Site Analysis

Location : Highstreet west

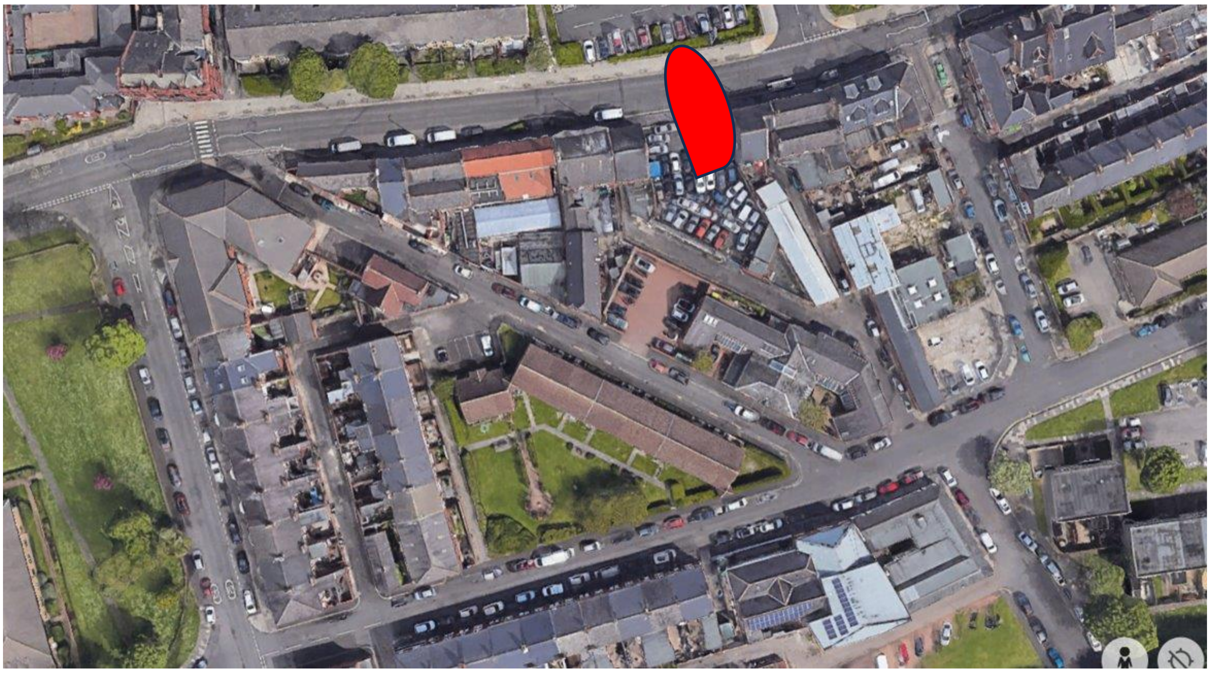


Fig 9- location of site

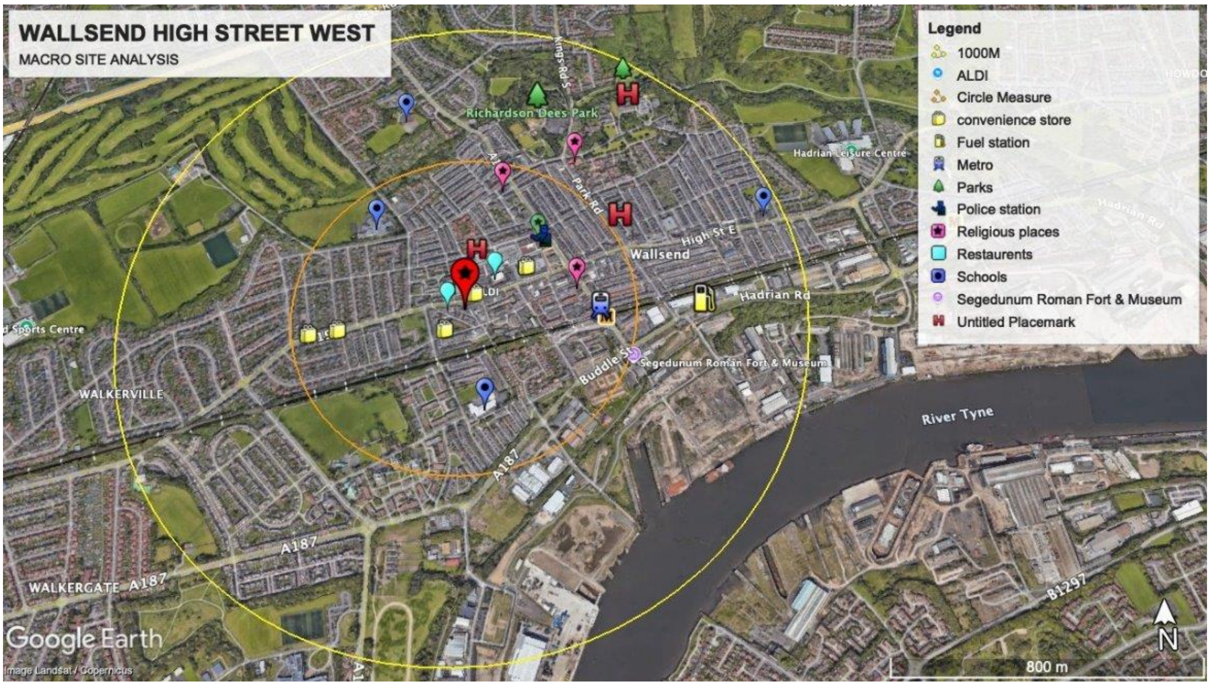


Fig 10 – macrolevel site analysis with a radius of 500m & 1000m

fig 11 . connectivity to site, major and minor roads with the site marked in red

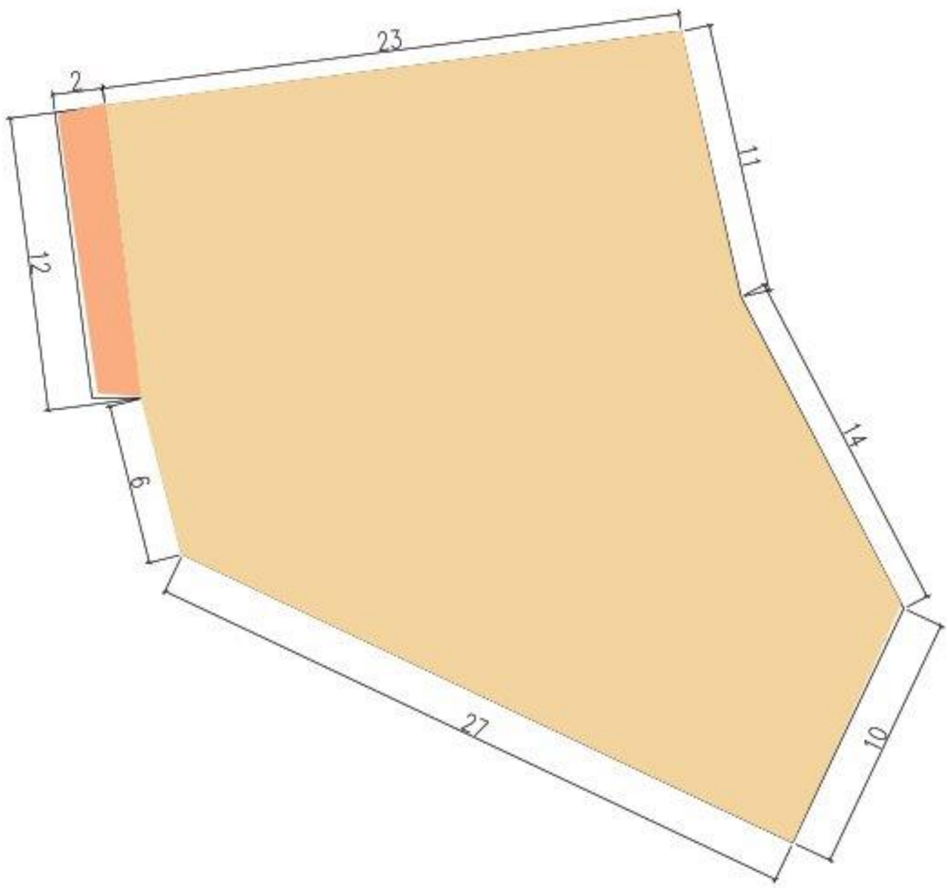


Fig 12 measured site dimensions

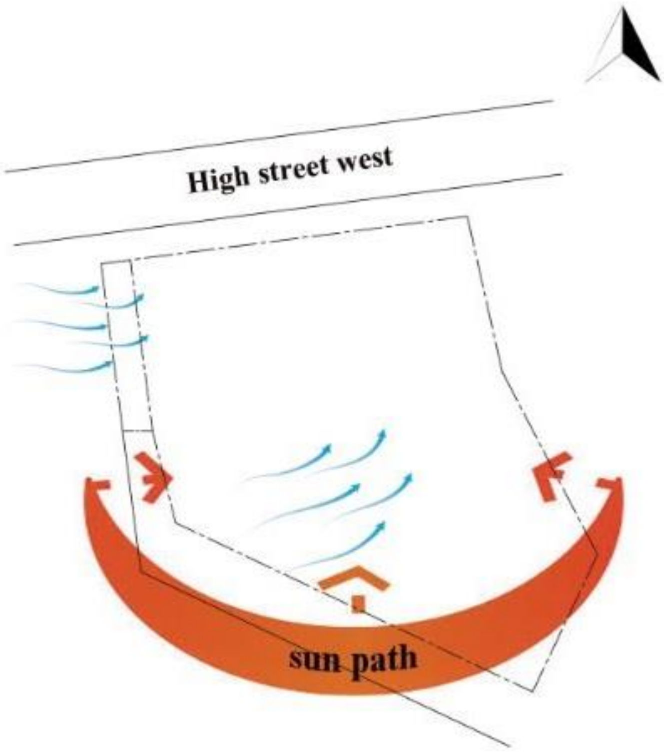


fig 13 sun path and wind direction in the site

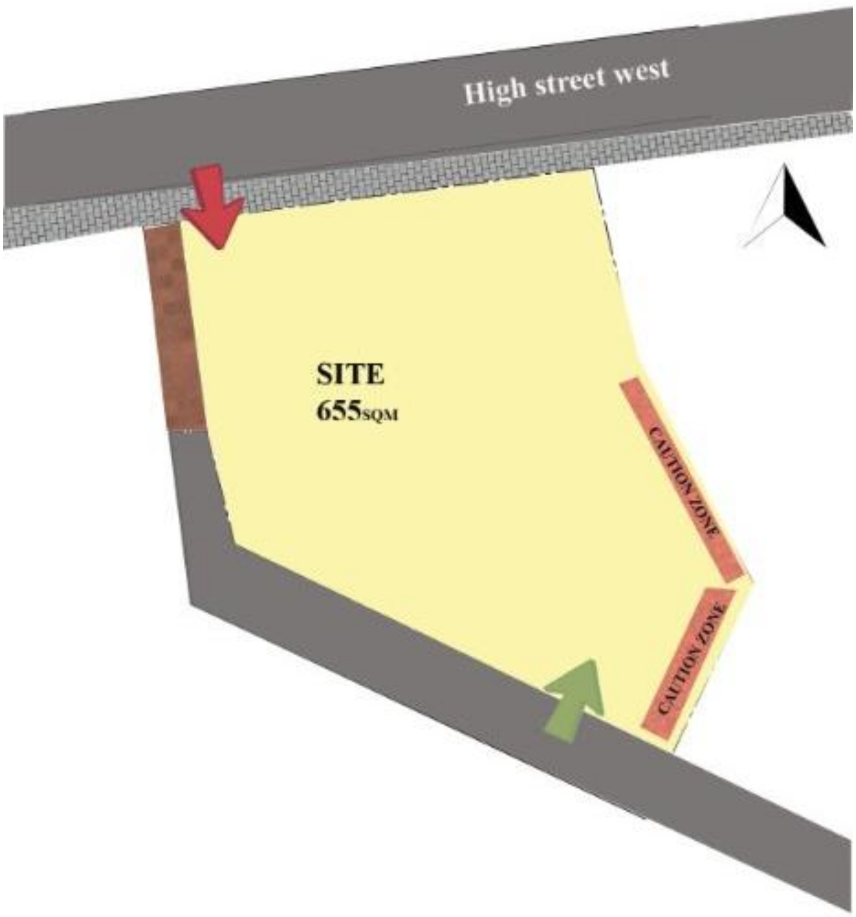


fig 14 access points and caution zone

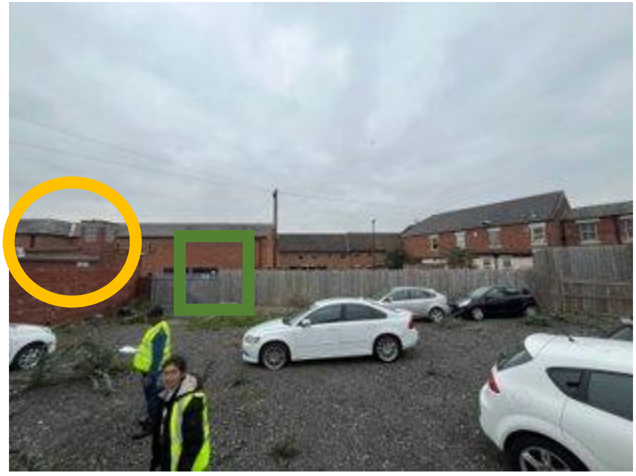


Fig 15 – group of pictures showing the real time access points and caution zone mentioned in fig 14

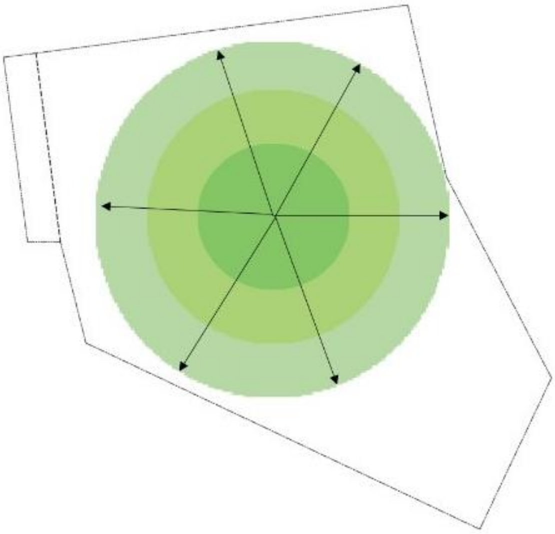
4. Design and Technical Specifications

4.1 Design Overview:

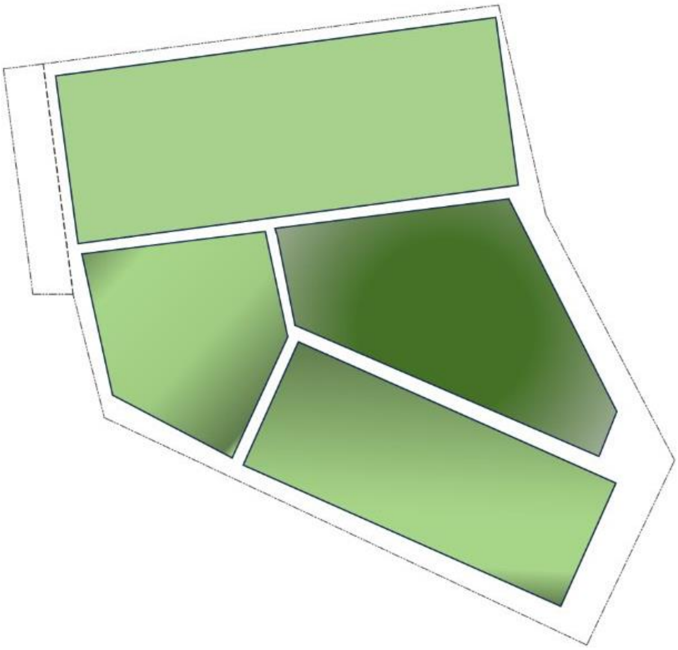
Concept :

- Ripple Effect Design: Layout inspired by ripple effect, expanding outward from a central point.
  - Core Focus: Communal garden and open spaces at the center for community bonding.
  - Community Engagement: Encourages connection, shared activities, and social interactions.
  - Inclusivity: Welcoming spaces for learning, teaching, and fostering relationships.
  - Balanced Living: Combines openness with privacy and security for residents.
  - Harmonious Environment: Promotes thriving communal and individual well-being.

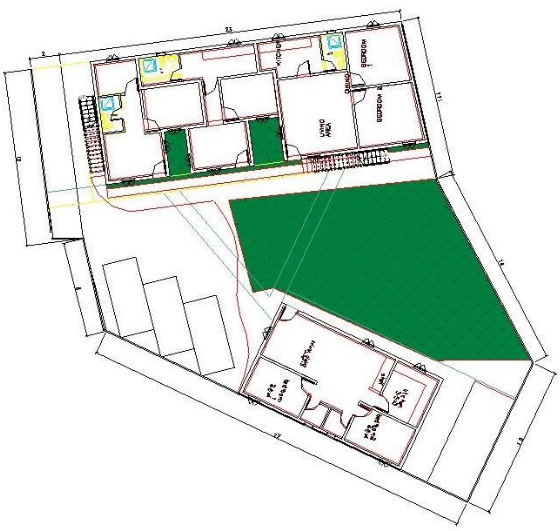
Fig 16. Conceptual images of design development



Ripple effect concept with the main focal point as the communal space



It is then broken to zones to accommodate different use of spaces



Development of conceptual plan

Fig 17 phases of design development from concept phase

Fig 18 Master plan

LEGENDS:

1. Residential building
2. Service building
3. Green communal space
4. Parking & alternative communal space
5. Public access pathway/ site access

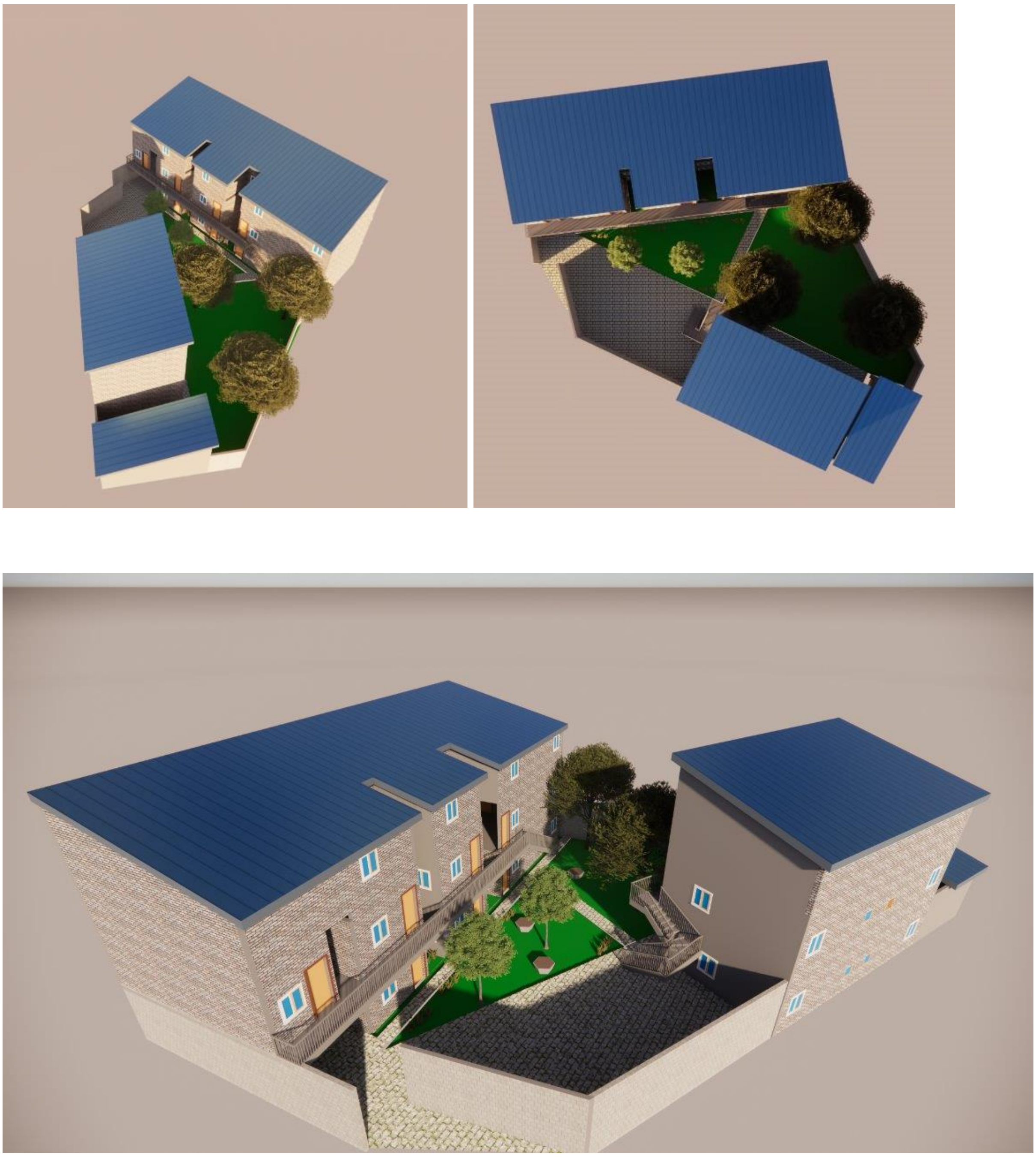


Fig 19. Ariel view of the masterplan

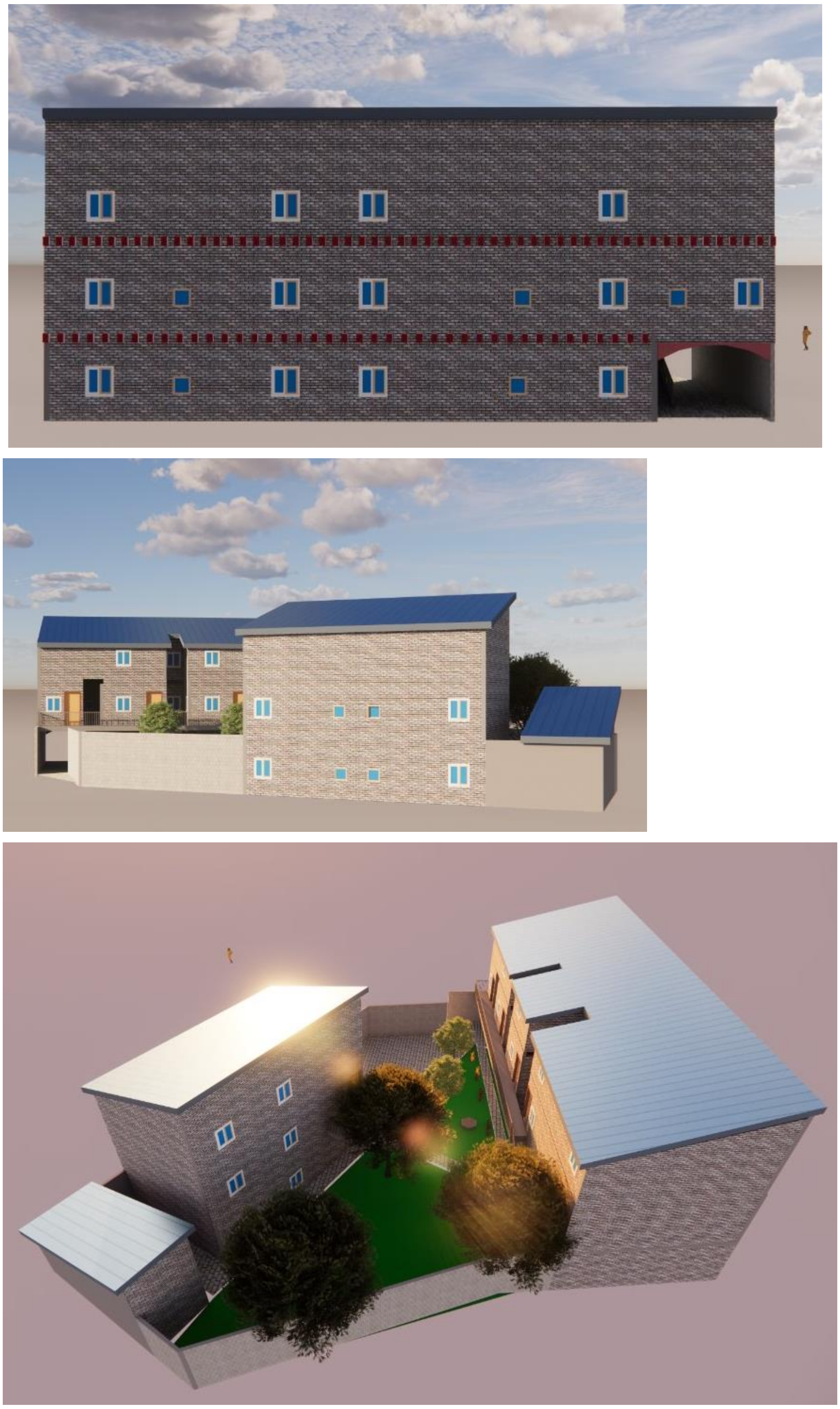


Fig 20. Views from different directions

Fig 21. Floor plans

4.2 Sustainable Materials for Net-Zero Housing Construction

Introduction

Sustainable construction is at the forefront of addressing the environmental challenges posed by traditional building practices. This report focuses on sustainable materials essential for constructing net-zero housing in Wallsend, Newcastle upon Tyne, UK. Four primary materials—hempcrete, cross-laminated timber (CLT), structural insulated panels (SIPs), and self-healing concrete—are complemented by additional materials such as mycelium, Ferrock, recycled steel, iron, and bricks to ensure a comprehensive approach to sustainable construction.

Hempcrete

Hempcrete, a biocomposite material made from hemp shives, lime, and water, is widely recognized for its environmental and structural benefits.

Properties and Benefits:

- Thermal Insulation: Hempcrete's low thermal conductivity (approximately 0.06 W/m·K) significantly enhances the energy efficiency of buildings.

- Carbon Sequestration: The cultivation of hemp absorbs CO_2, and the material continues to sequester carbon throughout its lifecycle.

- Lightweight: Its low density reduces the load on structural elements, making it ideal for walls and infill applications.

- Moisture Regulation: Hempcrete's hygroscopic properties regulate indoor humidity, contributing to occupant comfort.

Application:

Hempcrete is used primarily in wall construction, either as a cast-in-place material or in pre-formed blocks. For the Wallsend project, its integration will provide a sustainable alternative to traditional masonry while enhancing thermal performance.

Cross-Laminated Timber (CLT)

CLT is an engineered wood product made by bonding layers of timber at right angles, creating a robust and versatile construction material.

Properties and Benefits:

- Sustainability: Sourced from sustainably managed forests, CLT has a low embodied carbon footprint.

- Strength and Durability: Its high strength-to-weight ratio makes it suitable for load-bearing walls and floors.

- Ease of Construction: Prefabricated panels simplify assembly, reducing construction time and waste.

- Thermal Performance: With a thermal conductivity of approximately 0.12 W/m·K, CLT contributes to energy efficiency.

Application:

CLT will be used for structural elements such as walls, floors, and roofs. Its prefabrication aligns with the goals of precision and waste minimization in sustainable construction.

Structural Insulated Panels (SIPs)

SIPs are prefabricated panels consisting of an insulating core sandwiched between two structural facings, typically oriented strand board (OSB).

Properties and Benefits:

- Energy Efficiency: With U-values as low as 0.13 W/m^{2}K, SIPs minimize heat loss, contributing to net-zero energy goals.

- Structural Integrity: SIPs provide strength comparable to conventional framing while reducing material usage.

- Air Tightness: Their precision manufacturing ensures minimal gaps, enhancing airtightness and reducing energy consumption.

Application:

SIPs will be utilized for walls, roofs, and floors, providing a high-performance envelope for the net-zero housing units.

Self-Healing Concrete

Self-healing concrete incorporates bacteria or other agents to autonomously repair cracks, extending the material’s lifespan and reducing maintenance.

Properties and Benefits:

- Durability: The self-healing process mitigates crack propagation, enhancing structural integrity.

- Reduced Maintenance: By addressing micro-cracks early, it reduces the need for costly repairs.

- Sustainability: Fewer repairs result in lower carbon emissions over the structure’s lifecycle.

Application:

Self-healing concrete will be employed in foundations and load-bearing structures where durability and longevity are critical.

Iron

Iron, commonly used in construction, is an essential material for reinforcement and structural integrity.

Properties and Benefits:

- Structural Support: Iron’s high tensile strength makes it ideal for reinforcing concrete and load-bearing components.

- Durability: Properly treated iron resists corrosion and ensures long-term performance.

- Versatility: It is widely available and compatible with sustainable construction practices.

Application:

In the Wallsend project, iron will be used for reinforcements and structural support, contributing to the overall durability and safety of the housing units.

Bricks

Bricks have been a traditional construction material for centuries and continue to play a vital role in sustainable building.

<u>Properties and Benefits:</u>

- Thermal Mass: Bricks store heat during the day and release it at night, stabilizing indoor temperatures.

- Durability: High compressive strength ensures longevity and resistance to environmental stress.

- Recyclability: Used bricks can be repurposed, reducing waste.

<u>Application:</u>

For the Wallsend project, locally sourced bricks will be used in non-structural walls and cladding, aligning with sustainability principles by minimizing transportation emissions.

Conclusion

The selection of sustainable materials is critical to achieving the net-zero energy objectives for the Wallsend housing project. Hempcrete, CLT, SIPs, self-healing concrete, mycelium, Ferrock, recycled steel, iron, and bricks form a comprehensive material strategy. Together, these materials address key aspects of sustainability, including energy efficiency, carbon reduction, durability, and resource conservation. By integrating these innovative materials, the project not only fulfills its sustainability goals but also sets a benchmark for future net-zero housing developments.

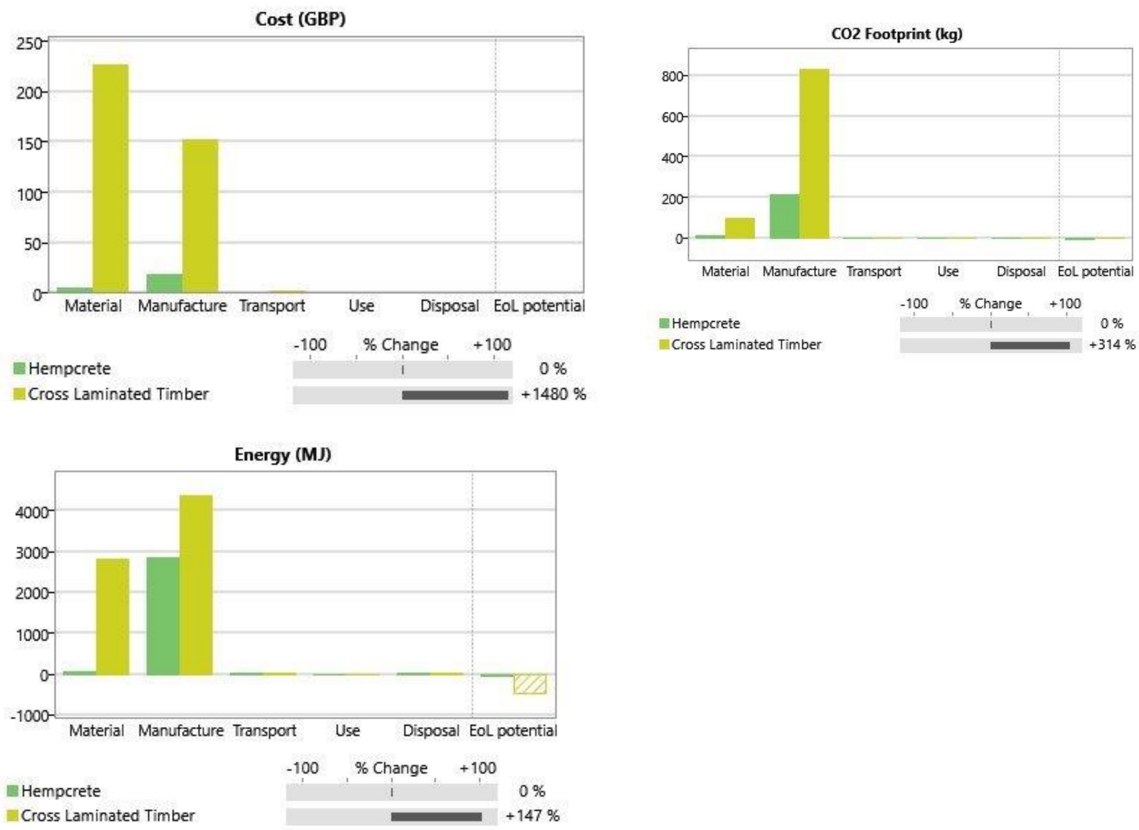


Fig 22. Eco audit chart of chosen material

4.3 Vendor Selection Summary for Construction Materials

Structural Insulated Panels (SIPs)

1. **Maitland Cool Rooms (MCR), Newcastle, NSW**
    - **Specialization**: Manufacturing, delivering, and installing EPS insulated panels for structural applications.
    - **Services**: End-to-end project management, from planning to on-site execution.
    - **Applications**: Clean rooms, morgues, wineries, private wine cellars, and more.
    - **Contact Information**:
        - Address: Unit 10/172 Racecourse Road, Rutherford, NSW 2320
        - Phone: Mick: 0402 690 607, Nathan: 0425 247 455
2. **Hemsec Manufacturing, UK**
    - **Specialization**: High-quality SIP panels and steel-faced insulated panels for residential, commercial, and industrial use.
    - **Key Features**: Sustainability-focused, PEFC-certified, and high-performing panels suitable for social housing, cold storage, and vertical farming.
    - **Contact Information**:
        - Address: Rainhill Head Office, Stoney Lane, Rainhill, Prescot, Merseyside, L35 9LL
        - Phone: 0151 426 7171

Self-Healing Concrete

1. **Breedon Newcastle Upon Tyne (Howdon) Concrete Plant**
    - **Specialization**: Ready-mixed concrete tailored for durability and long-term sustainability.
    - **Contact Information**:
        - Address: Howdon Concrete Plant, Willington Quay, Wallsend, Newcastle Upon Tyne NE28 6UR
        - Phone: 0191 615 3378
2. **Tarmac, UK**
    - **Specialization**: Sustainable construction solutions, including advanced concrete and asphalt products.
    - **Innovative Features**: Focus on decarbonization and next-generation construction techniques.
    - **Notable Projects**: HS2, St. Paul’s Cathedral restoration, and Wembley Stadium.

- **Contact Information**: Website: Tarmac

Cross-Laminated Timber (CLT)

1. **BSW Timber**
    - **Specialization**: Engineered wood products, including CLT, for prefabricated walls, ceilings, and roofs.
    - **Notable Projects**:
        - Stadthaus, London: First all-CLT building.
        - Mistissini Bridge, Canada: Award-winning CLT bridge.
    - **Contact Information**:
        - Phone: 0800 587 8887
        - Email: sales@bsw.co.uk
2. **Prudhoe Timber (Newburn) Ltd**
    - **Specialization**: Timber merchant offering high-quality materials for custom construction projects.
    - **Contact Information**:
        - Address: Bridgefields Peth Lane, Newburn Bridge Road, Newcastle upon Tyne, NE15 8NR
        - Phone: +44 7845 947502

General Timber Supplies

Quay Timber, Newcastle Upon Tyne

- **Specialization**: Wide range of timber products, including fencing, decking, and sheet materials.
- **Recent Projects**: Development in Ouseburn Valley Conservation Area, showcasing expertise in heritage-sensitive designs.
- **Contact Information**:
    - Address: Unit 5 Chillingham Industrial Estate, Back Chapman Street, Newcastle upon Tyne, NE6 2XX
    - Phone: 0191 224 0494
    - Email: info@quaytimber.co.uk

Hempcrete and Bio-Based Materials

1. **UK Hempcrete**
    - **Specialization**: Consultancy, architectural design, and specification of structural and thermal envelopes using hempcrete.
    - **Key Features**:
        - Hempcrete provides stable internal temperatures and humidity with minimal mechanical and electrical load.
        - Demonstrated performance: Internal temperatures of 16–18°C year-round, regardless of external conditions.
    - **Contact Information**:
        - Additional insights available in their publication: "Can We Afford Hempcrete? An In-Depth Look at Hempcrete Construction Costs."

4.4 Renewable energy solutions:

In the pursuit of sustainable living and energy efficiency, incorporating renewable energy solutions into residential properties has become increasingly essential. For a house located in Wallsend, the transition to renewable energy presents a unique opportunity to reduce carbon emissions, lower energy costs, and contribute to the community's environmental goals. This proposal explores tailored solutions, including solar panels, energy-efficient roofing and windows, and seeking scope of VPP designed to harness renewable energy sources effectively while meeting the specific needs of the property and its occupants.

<u>DEMAND CALCULATION FOR AVERAGE 30 ppl ACCOMODATION</u>

Approximation for Single Accommodation:

Baseline electricity usage per person: Single person in a small apartment: ~1,500 to 2,500 kWh/year.

Electricity demand for 30 people: For 30 individuals: 30×1,50030 \times 1,50030×1,500 to 30×2,50030 \times 2,50030×2,500 = 45,000 to 75,000 kWh/year.

Monthly: 45,000/1245,000/1245,000/12 to 75,000/1275,000/1275,000/12 = 3,750 to 6,250 kWh/month.

Daily electricity demand: Assuming 30 days per month: 3,750/303,750/303,750/30 to 6,250/306,250/306,250/30 = 125 to 208 kWh/day.

Specific Scenarios:

Low electricity use (energy-efficient appliances): ~1,200 kWh/year per person, total ~36,000 kWh/year for 30 people.

High electricity use (frequent heating/cooling): ~3,000 kWh/year per person, total ~90,000 kWh/year for 30 people.

<u>CALCULATION:</u>

To calculate the area of solar panels required to generate 45,000 kWh/year in the UK, we need to consider:

1. Solar panel efficiency: Most modern solar panels have an efficiency of 15-20%.
2. Average solar energy in the UK: The UK receives about 1,000 kWh of solar energy per square meter per year on average.
3. Panel performance ratio: Accounting for losses due to temperature, shading, and system inefficiencies, the performance ratio is typically around 80-85% in the UK.

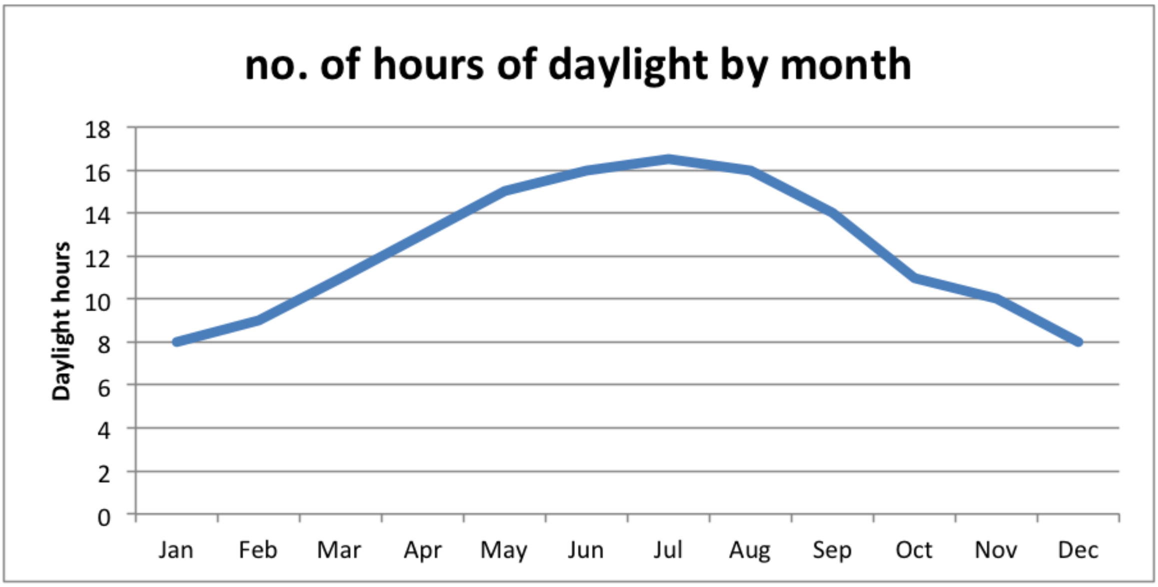


Fig 23

Required Solar Panel Area Calculation

1. Output per m² in the UK = Average: 127.5 kWh/year/m².
2. Area Required (45,000 kWh/year) =Total: 353 m² (approx.).
3. Installation Costs = £1,000-£1,500/kW → Total: £25,000 to £45,000.

<u>Considerations:</u>

Panel wattage: Modern panels are typically 300-400 W, requiring ~3-4 m² per kW.

Roof size and orientation: South-facing roofs are ideal in the UK; otherwise, efficiency may drop.

Battery storage: For consistent energy availability, a battery system may be needed.

We need approximately 350–360 m² of solar panel installation in the UK to generate 45,000 kWh/year. The cost of installing solar panels for a 365 m² roof with an inclined design will depend on the type of solar panels, system size, and installation complexity. Here's an estimate based on current UK prices:

1. System Size: For a large roof like 365 m², you could potentially install a system with a capacity of up to 25-30 kW. This size is suitable for significant energy generation, potentially for commercial or large residential properties.
2. Cost Estimate: The average cost for solar panels, including installation, is approximately £1,000-£1,500 per kW for larger systems. For a 25-30 kW system, the total cost could range between £25,000 and £45,000. This estimate includes panels, inverters, mounting systems, and labour but excludes optional components like battery storage.
3. Installation Complexity: An inclined roof may increase installation costs slightly due to additional labour or specialized mounting requirements. Labor costs in the UK typically range between £600 and £3,000, depending on the complexity and duration of the work. To address these challenges and ensure maximum efficiency, we designed the roofing with a single slant facing the south side to optimize exposure to sunlight

4. Government Incentives: Programs like the Smart Export Guarantee (SEG) can help you earn income from surplus energy exported to the grid, reducing the payback period for your investment.

Selection of solar panel for roof: In the UK, monocrystalline solar panels are generally the most suitable choice for residential and commercial installations due to their high efficiency, excellent performance in low-light conditions, and durability. These panels boast efficiency rates of 15–22%, making them ideal for the UK's less intense sunlight and variable weather. They also perform well on overcast days and require less roof space to generate significant energy, which is beneficial for smaller properties. Additionally, monocrystalline panels are known for their durability, often coming with longer warranties to ensure reliability in the UK’s variable climate. While polycrystalline solar panels are more affordable and suitable for properties with ample roof space, they are less efficient (13–16%) and perform poorly in low light compared to monocrystalline options. Thin-film solar panels, although lightweight and flexible, have low efficiency (10–12%) and are better suited for niche applications. Therefore, for most UK homeowners, monocrystalline solar panels offer the best combination of efficiency, durability, and space-saving benefits, ensuring consistent energy production even in less favourable weather conditions.

Selection of suitable solar panel for windows: Transparent solar panels, or photovoltaic glass, combine energy generation with functionality, allowing natural light while producing electricity. Ideal for modern buildings, they blend seamlessly into windows, preserving aesthetics while reducing energy costs and carbon footprints. These panels maximize energy use without additional space, offering thermal insulation that reduces heat loss in winter and heat gain in summer. They are versatile, suitable for homes, offices, greenhouses, and vehicles, and provide natural lighting with reduced glare, enhancing indoor comfort. Transparent solar panels align with smart city goals by integrating renewable energy into infrastructure, promoting sustainability, and improving energy efficiency without compromising design.

Best brands for solar panels in UK :

1. SunPower (Maxeon Solar Technologies).

2. LG Solar (NeON Series)

3. Panasonic (EverVolt HIT Series)

4. REC Solar

5. Canadian Solar (HiKu and HiDM Series)

6. JA Solar

7. Trina Solar

8. JinkoSolar

9. Q CELLS

Solar installation and maintenance team in North east UK:

GeoWarmth
GeoWarmth specializes in solar panel installations, battery storage, and EV charging for homes and businesses. With over 20 years of experience in sustainable energy, they offer tailored renewable energy solutions designed for optimal performance, even in cloudy conditions

SkylarSolar
Skylar Solar provides bespoke solar energy systems in Newcastle, offering services from design to installation and aftercare. Their focus on advanced technology integration ensures high-efficiency systems tailored to Newcastle's weather and environmental conditions

North East Solar Panels: This company offers solar panel installation and maintenance services with a focus on affordability and reliability. They provide energy solutions for homes and businesses in the Newcastle area

SCOPE OF AIR SOURCE HEAT PUMP IN UK :

Air source heat pumps (ASHPs) have significant potential in the UK, especially as the country works towards its goal of net-zero carbon emissions by 2050. The government is actively promoting ASHP adoption through incentives like the Boiler Upgrade Scheme, offering grants up to £5,000 to replace gas boilers with low-carbon alternatives. ASHPs are highly efficient, even in the UK's moderate climate, extracting heat from the air year-round, making them suitable for most regions. They offer a cost-effective solution for heating homes, especially with rising energy costs, and the lower running costs help offset the high initial installation price over time. ASHPs are also well-suited for both new builds and retrofitting existing homes, contributing to their growing popularity. Despite challenges like installation costs and the need for well-insulated homes, technological advancements and government support are expanding their scope, positioning ASHPs as a key part of the UK's sustainable energy future.

As of 2024, there has been significant growth in the installation of air source heat pumps (ASHP) across the UK. In the first six months of 2024 alone, approximately 26,797 certified heat pump installations were completed, reflecting a 45% increase compared to the same period in 2023. This surge indicates the growing trend towards renewable heating solutions as the UK moves towards a net-zero future. The UK government has set ambitious targets, aiming to install 600,000 heat pumps annually by 2028, further emphasizing the importance of this technology in reducing carbon emissions and enhancing energy efficiency

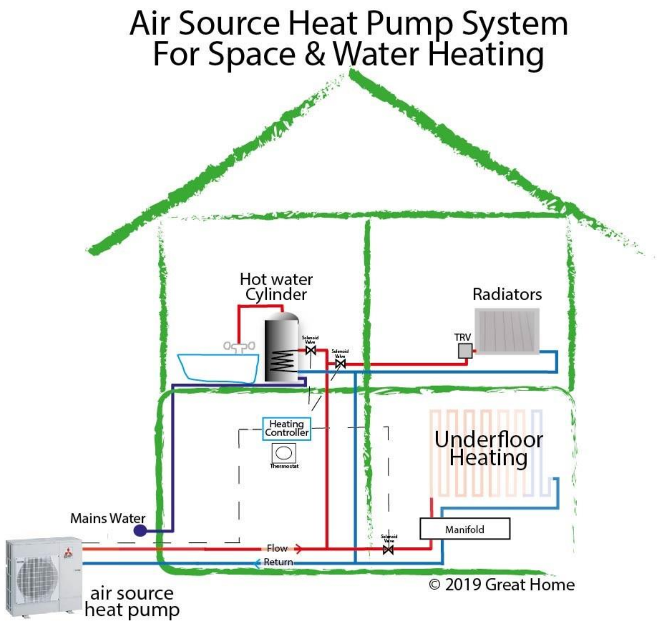
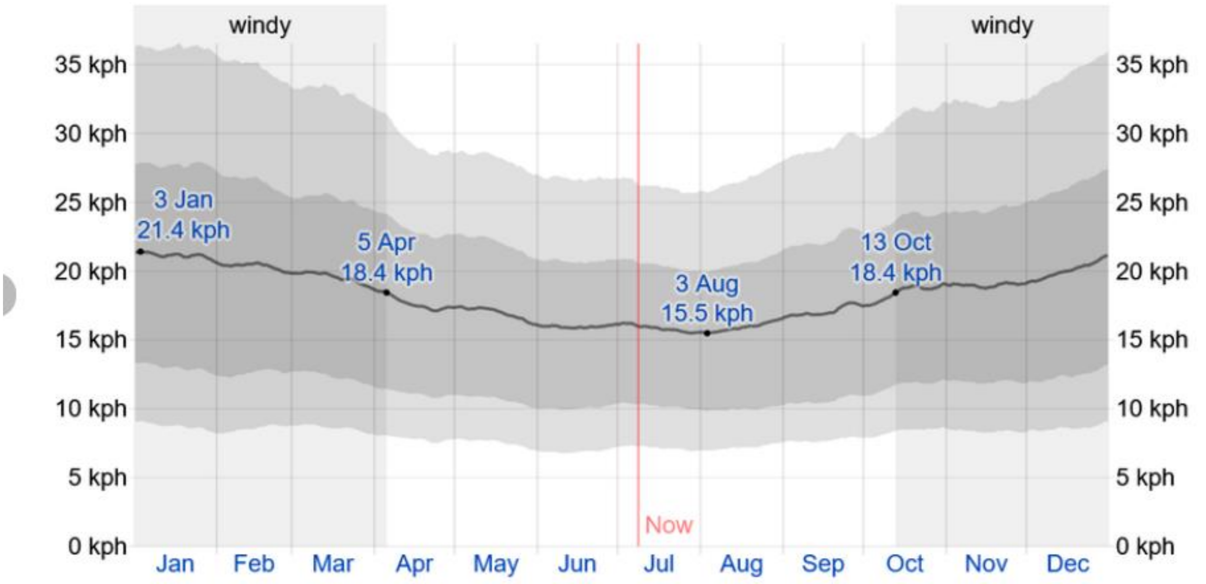


Fig 24

AVERAGE WIND SPEED IN UK



(a) shows the yearly wind direction in the UK (b) yearly average wind speed in the UK [42].

Fig 25

- Estimated cost of installation £12,000-£15,000 (Considering 26 people accommodation)

Air source heat pumps (ASHPs) offer several advantages for homeowners and businesses in the UK, particularly as part of the country's push toward renewable energy solutions.

1. Energy Efficiency

ASHPs are highly efficient systems, capable of producing multiple units of heat for every unit of electricity consumed. They typically operate at efficiencies of 300-400%, which means they are more energy-efficient than traditional heating systems like boilers. This leads to lower energy bills and reduced carbon emissions, making them an attractive option for environmentally conscious consumers.

2. Reduced Carbon Footprint

Since ASHPs use renewable heat from the air, they reduce reliance on fossil fuels. This can significantly cut a property's carbon emissions, supporting the UK's commitment to reaching net-zero emissions by 2050.

3. Lower Operating Costs

While the initial installation cost can be higher compared to traditional heating systems, ASHPs have lower ongoing running costs. Homeowners can save on energy bills, particularly when replacing outdated oil or gas heating systems.

4. Versatility and Easy Integration

ASHPs can be used for both heating and cooling, making them suitable for year-round comfort. They are also easier to integrate into existing homes compared to other systems, such as ground-source heat pumps, which require extensive groundworks.

5. Government Incentives

The UK government supports the installation of ASHPs through grants like the Boiler Upgrade Scheme, which can cover up to £7,500 of installation costs. This makes transitioning to an air source heat pump more affordable.

6. Longevity and Low Maintenance

ASHPs typically have a lifespan of around 20-25 years and require minimal maintenance compared to traditional heating systems. Annual servicing ensures optimal performance and helps extend the unit's longevity.

BRANDS OF AIR SOURCE HEAT PUMP IN UK :

Worcester Bosch offers the Compress 7001iAW, a popular choice for its ease of maintenance, quiet operation (55dB), and high efficiency. It's ideal for medium-sized properties and has an excellent customer support reputation

Samsung's EHS R32 Monobloc heat pump is renowned for its energy efficiency, with an output of up to 16kW, and can be connected to smart home systems. It operates at a low noise level of 54dB

Panasonic's Aquarea Monobloc is a sleek, modern option, best for well-insulated smaller homes, offering a heat output of up to 7kW

VENDORS AND INSTALLATION COMPANY's IN UK :

- Grant UK - Known for their Aerona 290 series, Grant offers energy-efficient heat pumps with low noise levels and a long warranty period of up to 7 years when installed by accredited engineers

Vaillant - This company provides versatile and high-efficiency heat pumps like the aroTHERM plus, which combines with water cylinders to ensure excellent performance for homes with multiple bathrooms

Daikin - Daikin's Altherma series is renowned for its versatility, providing hybrid solutions that combine heat pumps with existing boilers. The company's models are also compatible with solar panel systems, enhancing energy efficiency

Hitachi - Hitachi's Yutaki range is known for its cost-saving capabilities, with models that offer both heating and cooling functions, suitable for different home setups

Virtual Power Plant (VPP): Revolutionizing Energy in the UK

A Virtual Power Plant (VPP) in the UK is a digitally managed network of decentralized energy resources like solar panels, wind turbines, batteries, and flexible demand assets (e.g., EV chargers). VPPs aggregate and optimize these resources to act as a single power plant, providing energy to the grid, participating in markets (e.g., wholesale energy, capacity, and balancing), and delivering grid services like frequency response and load shifting. By leveraging IoT, AI, and advanced software, VPPs enhance grid stability, enable efficient renewable energy use, and support the UK's Net Zero by 2050 target. Key players like Octopus Energy and Centrica are advancing VPP adoption across the UK.

The amount earned from selling energy to a Virtual Power Plant (VPP) in the UK depends on factors such as the type of energy source, market conditions, and the specific VPP program. Participants can receive payments through mechanisms like the Smart Export Guarantee (SEG), which allows energy producers to sell surplus electricity to the grid at rates determined by energy suppliers. Rates can range between 5p to 15p per kWh, depending on the supplier and contract terms. For specialized VPP programs like Octopus Energy's "Intelligent

Octopus," customers can earn competitive rates, with costs for clean electricity as low as 10p per kWh during periods favourable for the grid. These programs often combine savings with earnings by optimizing the timing of energy export and usage to balance the grid effectively

If you have solar panels, battery storage, or other distributed energy resources, participation in a VPP could provide additional financial incentives while promoting renewable energy integration. It's worth consulting specific VPP providers to understand their payment structures and benefits fully.

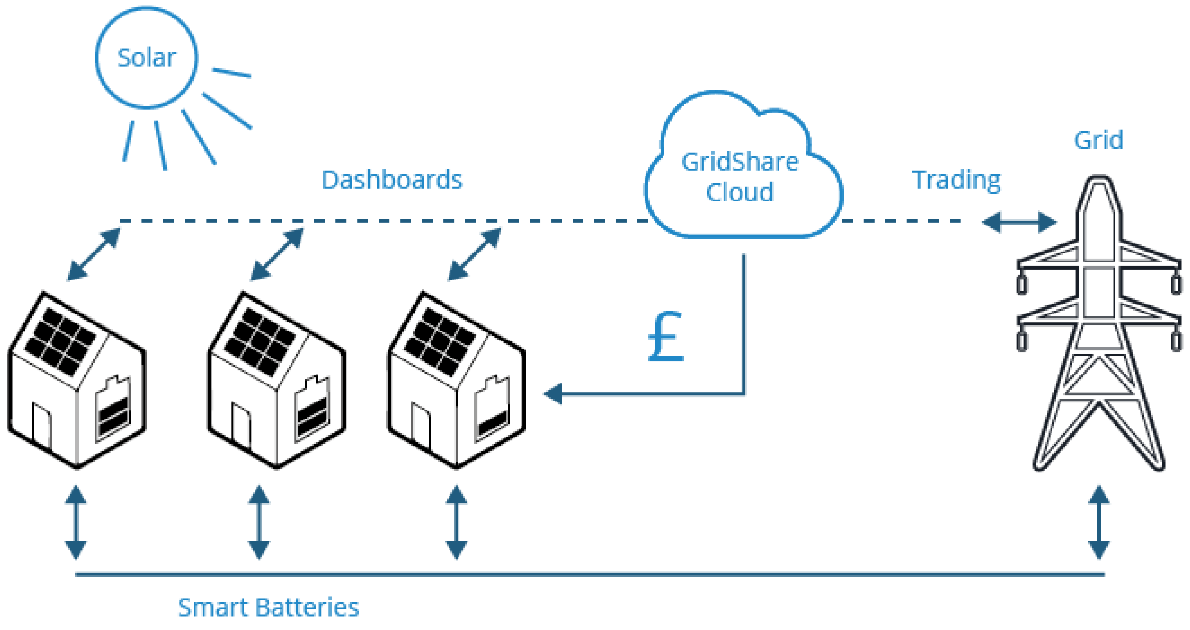


Fig 26

Connecting a solar panel system to a virtual power plant (VPP) involves integrating renewable energy systems into a network that optimizes and trades energy. Here's a step-by-step process:

1. Set Up the Renewable Energy Systems

- Install the Solar Panels: Ensure proper installation by certified professionals. Each system must meet local regulations and safety standards.
- Inverters: Both systems require inverters to convert generated DC electricity into AC for grid compatibility. Hybrid inverters can handle multiple energy sources like wind and solar.

2. Install Energy Management Systems (EMS)

- Smart Metering: A smart meter monitors energy generation, consumption, and export. This data is essential for VPP participation.
- EMS Software: Install software or hardware that communicates with the VPP, enabling real-time energy monitoring and control. Many systems are cloud-based for remote access.

3. Battery Storage Integration (Optional)

- Adding battery storage allows energy generated by your systems to be stored and dispatched to the grid or used during peak times, enhancing your contribution to the VPP.

4. Connect to the VPP

- Choose a VPP Provider: Select a VPP operator or aggregator in your region that supports renewable energy assets (e.g., Octopus Energy in the UK).
- Register Your Assets: Provide technical details about your wind turbine, solar panels, and battery storage (if applicable) to the VPP operator.
- Install Communication Hardware: Some VPPs require proprietary communication devices or integrations to connect your system to their platform.

5. Grid Compliance

- Ensure that your renewable systems comply with grid connection standards, such as G99 regulations in the UK for generators over 16A per phase.
- Work with a qualified electrician to ensure all connections are safe and approved by the local Distribution Network Operator (DNO).

6. Energy Export and Optimization

- Once connected, your renewable energy systems can export excess energy to the VPP, which optimizes distribution based on demand.
- The VPP may offer compensation based on real-time energy trading, benefiting from energy generated during peak times.

7. Monitor and Maintain

- Use the VPP’s dashboard or app to track your system’s performance, energy contribution, and earnings.
- Regular maintenance ensures that your wind turbine and solar panels operate efficiently.

Additional Considerations

- Licensing: Depending on the size of your system, you may need generation licenses or approval from your local electricity authority.
- Contracts: Review your agreement with the VPP provider to understand payment rates and responsibilities.

INTERFACE OF EMS SOFTWARE

To connect an Energy Management System (EMS) in your home, you'll need to integrate various devices like smart meters, appliances, energy storage systems, and renewable energy sources. Here's how you can set up an EMS and examples of systems you can use:

Steps to Connect EMS at Home

1. Assess Your Home Energy Setup:
   - Identify energy sources (solar panels, grid electricity, batteries).
   - Determine the smart appliances and meters already in use.
2. Choose an EMS Platform:
   - Select an EMS that matches your needs, such as monitoring energy usage, integrating solar panels, or optimizing battery use.
3. Install Smart Meters and IoT Devices:
   - Smart meters measure energy consumption and production.
   - IoT devices connect appliances to the EMS for real-time data sharing.
4. Connect Renewable Sources and Storage:
   - Link your solar inverters or wind turbines and battery storage systems to the EMS.
5. Set Up a Communication Gateway:

- Use Wi-Fi, Zigbee, or proprietary protocols to ensure devices communicate with the EMS.

6. Configure and Monitor:

- Use the EMS interface (app or web platform) to configure preferences, monitor energy flows, and schedule usage.

Examples of EMS for Home Use

1. Tesla Powerwall

- Manages solar energy and battery storage, optimizing grid interaction.
- Features include blackout backup and real-time monitoring via the Tesla app.

2. Sonnen EcoLinx

- Integrates with home automation systems to optimize energy use and reduce costs.

3. Schneider Electric Wiser Energy

- Tracks energy use in real-time and provides actionable insights.

4. Enphase Energy System

- Designed for homes with solar panels, it offers detailed energy flow monitoring and battery integration.

5. Siemens Desigo

- A flexible EMS that supports renewable energy integration for residential and commercial applications.

6. Hive Active Heating and Home Energy Management

- Helps optimize heating, cooling, and appliance energy use while connecting to renewable sources.

Benefits of EMS at Home

- Reduced energy bills through optimized energy usage.
- Enhanced energy independence with integrated solar and batteries.
- Real-time insights for smarter energy consumption decisions.

Water Conservation

Water is a precious and finite resource, facing increasing pressure from population growth and climate change. Conservation is crucial for ensuring sustainable water supplies for future generations. Sustainable water management practices are essential for safeguarding our planet's well-being. By embracing water conservation, we can protect our environment and ensure a thriving future.

Wallsend is a city with a temperate climate, characterized by warm temperatures and significant rainfall throughout the year. Even during its driest months, precipitation remains substantial. With an annual rainfall of 718 mm (28.3 inches), Wallsend presents ideal conditions for rainwater harvesting.

Rain Water Estimated Calculation

Key Data:

1. Catchment Area: 281 m^2
2. Annual Rainfall: 718 mm (0.718 m)
3. Runoff Coefficient: 0.9

Calculation:

Harvested Water = $281 \times 718 \times 0.9$

= 201,783.29 litres/year

Monthly Potential:

Since rainfall is fairly evenly distributed:

Monthly Harvest = 201,783.29/12 ≈16,815 liters/month

Regulations and Standards for Water Conservation

1. Building Regulations

Part G: Sanitation, Hot Water Safety, and Water Efficiency

Ensure potable water consumption does not exceed 125 liters per person per day

Use water-efficient fixtures like low-flow taps, dual-flush toilets, and aerated showerheads.

Include water meters for monitoring consumption.

Part H: Drainage and Waste Disposal

Rainwater and greywater systems must be connected to appropriate drainage systems to prevent contamination.

2. Sustainable Drainage Systems (SuDS) Compliance

Guidance: Align with the Flood and Water Management Act 2010 and local council regulations.

Install SuDS such as permeable paving, infiltration systems, and green roofs to manage rainwater.

Prevent stormwater runoff from overloading local drainage networks.

3. BREEAM Standards for Water Management

Purpose: Maximize water efficiency and reduce wastage in sustainable building projects.

Reduce water consumption compared to a baseline building through water-efficient fixtures and systems.

Implement rainwater harvesting and greywater recycling systems.

Monitor water usage with leak detection and sub-metering for high-demand areas.

Overview of UK Government Rainwater Harvesting Programs

The UK government recognizes the environmental and economic benefits of rainwater harvesting, and has implemented several initiatives to promote its adoption. These programs aim to reduce reliance on potable water, conserve water resources, and mitigate the impact of climate change. Key programs include:

- The Water Efficiency Scheme (WES): This program offers grants and incentives for a wide range of water-saving technologies, including rainwater harvesting systems.
- The Green Homes Grant Scheme: This scheme provides grants to homeowners for energy efficiency improvements, which can include installing rainwater harvesting systems.
- Local Council Incentives: Many local councils in England also offer grants and incentives specifically for rainwater harvesting, often tailored to the needs of their communities.

Maintenance and Operational Expenses

1.Tank Cleaning

Regular tank cleaning ensures water quality and prevents buildup of sediment.

2.Filter Replacement

Filters need to be replaced periodically to maintain water purity.

3.Pump Maintenance

Regular pump inspections and maintenance prevent malfunctions and ensure efficient operation.

Filtration Options:

1. Pre-filtration: Removes larger particles such as leaves and sand before water enters the storage tank. Common examples are mesh filters and vortex separators.
2. Sediment Filtration: Targets finer particles suspended in water, such as dirt or silt. Examples include cartridge filters or sand filters.
3. Activated Carbon Filtration: Removes chlorine, organic contaminants, and odors, improving water quality for non-potable uses like irrigation or flushing.

CONCLUSION

The integration of sustainable water and energy management solutions for a property in Wallsend represents a comprehensive approach to environmental and economic sustainability. By capitalizing on Wallsend's temperate climate and substantial rainfall, rainwater harvesting systems can reduce potable water consumption and manage stormwater efficiently, complementing compliance with building regulations, Sustainable Drainage Systems (SuDS), and BREEAM standards. These water conservation efforts align with government programs, such as the Water Efficiency Scheme and Green Homes Grant Scheme, to promote sustainable practices.

Simultaneously, transitioning to renewable energy solutions, including solar panels and Air Source Heat Pumps (ASHP), delivers significant environmental and financial benefits. The incorporation of a Virtual Power Plant (VPP) and Energy Management Systems (EMS) enhances energy efficiency, optimizes consumption, and provides opportunities to earn from surplus energy, contributing to the UK's net-zero targets. Government incentives, such as the Smart Export Guarantee (SEG) and Boiler Upgrade Scheme, further support the financial viability of adopting these technologies.

Together, these initiatives demonstrate how integrated water conservation and renewable energy strategies can achieve reduced carbon emissions, lower long-term costs, and enhance environmental resilience. Wallsend can serve as a model for sustainable urban development, offering scalable solutions that balance ecological responsibility with economic practicality.

5. Conclusion and Recommendations

5.1 Summary of Findings

The analysis presented in this report underscores the critical role of sustainable construction materials and practices in addressing modern challenges of urban development. Key findings include:

- **Sustainable Materials**: Structural Insulated Panels (SIPs), self-healing concrete, Cross-Laminated Timber (CLT), and hempcrete offer significant environmental and economic benefits. These materials improve energy efficiency, enhance structural durability, and reduce carbon emissions.
- **Vendor Expertise**: Vendors such as Maitland Cool Rooms, Hemsec Manufacturing, UK Hempcrete, Tarmac, and BSW Timber provide tailored solutions for various construction applications, from residential buildings to large-scale infrastructure projects.
- **Renewable Energy and Water Conservation**: Integrating solar panels, Air Source Heat Pumps (ASHP), Virtual Power Plants (VPP), and rainwater harvesting systems ensures energy and water efficiency. These measures align with net-zero goals and reduce long-term operational costs.
- **Regulatory Compliance**: Adherence to standards such as BREEAM, building regulations (Part G and H), and Sustainable Drainage Systems (SuDS) ensures environmentally responsible development and legal conformity.
- **Holistic Development**: Case studies demonstrate that a combination of advanced materials, innovative design, and stakeholder engagement fosters sustainable, affordable, and high-quality living environments.

5.2 Recommendations

Based on the findings, the following recommendations are proposed:

1. **Material Selection and Vendor Collaboration**:
   - Prioritize sustainable materials like SIPs, hempcrete, and CLT for their energy efficiency and durability.
   - Engage with specialized vendors, such as Hemsec Manufacturing for SIPs and UK Hempcrete for bio-based materials, to ensure tailored solutions that align with project goals.
2. **Renewable Energy Integration**:
   - Incorporate solar panels, ASHP, and VPP systems to reduce reliance on non-renewable energy sources and optimize energy efficiency.
   - Leverage government incentives like the Smart Export Guarantee (SEG) and Boiler Upgrade Scheme to improve the financial feasibility of these systems.
3. **Water Management Systems**:
   - Implement rainwater harvesting and greywater recycling systems to enhance water conservation and reduce strain on potable water supplies.
   - Design infrastructure in compliance with SuDS to manage stormwater and mitigate flooding risks.

4. **Regulatory and Stakeholder Engagement**:
   - Ensure all developments comply with relevant regulations, including building codes, SuDS guidelines, and BREEAM standards.
   - Foster collaboration with local authorities, residents, and industry stakeholders to align development goals with community needs.
5. **Adopt Case Study Best Practices**:
   - Learn from successful projects such as Goldsmith Street and Marmalade Lane to incorporate energy-efficient designs and communal living models.
   - Emulate innovative approaches like modular construction and bio-based material use for scalable and replicable solutions.
6. **Monitor and Optimize**:
   - Establish systems to monitor energy and water usage post-construction, ensuring performance goals are met.
   - Continuously evaluate and adapt strategies to incorporate emerging technologies and improve sustainability outcomes.

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